SS-S-111B
May 18, 1976
SUPERSEDING
Fed. Spec. SS-S-111A
January 12, 1968

FEDERAL SPECIFICATION

SOUND CONTROLLING MATERIALS (TROWEL AND SPRAY APPLICATIONS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, **for the use of** all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers acoustical materials for trowel or spray applications.

1.2 Classification. The acoustical materials shall be of the following types, densities, grades, and classes as specified (see 6.2):

Type I - Cementitious materials.
  Density A - 18 to 20 pounds per cubic foot (ppcf)
  Density B - Over 20 ppcf.

Type II - Fibrous materials.
  Density C - 11 to 14 ppcf.
  Density D - Over 14 ppcf.

Type III - Synthetic polymeric materials.
  Density E - 2 to 12 ppcf.
  Density F - Over 12 ppcf.

Grades 1 through 7 according to level of noise reduction required (see 3.5.1).
Grades 101 through 107 according to level of sound absorption required (see 3.5.2).

Class (a) - Noncombustible (see 3.6.1).
Class (b) - Fire-resistant (see 3.6.2).
Class (c) - Slow burning (see 3.6.3).

FSC 5610

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

SS-B-755 - Building Board, Asbestos-Cement: Flat and Corrugated.
UU-S-48 - Sacks, Shipping, Paper.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards, as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Dallas, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing And Materials (ASTM) **Standards:**

C423 - Sound Absorption of Acoustical Materials in Reverberation Rooms, Test for.
E84 - Surface Burning Characteristics of Building Materials, Test for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Certificate of compliance. Unless otherwise specified (see 6.2), acoustical material offered under this specification shall be certified by the contractor to have successfully completed the tests for sound absorption, flame spread index, and dust.

3.2 Standard commercial product. The acoustical material shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commerical product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commerical product shall be included in the acoustical material being furnished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

Standard commerical product is a product which has been or will be sold on the commerical market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production material(s).

3.3 First article. When specified (see 6.2), the contractor shall supply containers of acoustical material in quantities as specified (see 6.2), of each type, density, grade, and class of material as specified (see 6.2), for first article inspection and approval (see 4.2.1 and 6.3).

3.4 Material.

3.4.1 Type I. Type I ingredients shall be uniformly mixed and ready for use with the addition of water. When received, the material in the container shall be dry and shall not be caked or sticky. After it has been properly mixed with the amount of water specified by the manufacturer, the materials shall be of such a consistency that it can be applied with a plastering trowel or with a suitable plaster-spraying machine. Upon drying, the acoustical material shall have a hardness adequate to form a durable surface.

3.4.2 Type II. Type II shall consist of mineral or vegetable fiber with an integrally mixed binder, or it may be supplied for use with a liquid binder added at the time of application.

3.4.3 Type III. Type III shall consist of synthetic polymeric materials, two-component, rigid urethane spray foam. This product can be applied with foam dispensing equipment.

3.5 Acoustical properties.

3.5.1 Noise-reduction coefficient (NRC). Acoustical materials used to reduce the noise level shall be classified in terms of sound-absorbing qualities at four frequencies: 250, 500, 1000, and 2000 Hertz (Hz). The average of the sound-absorption coefficients at these four frequencies will be called the noise-reduction coefficient. For grading purposes, the average will be taken to the nearest multiple of 0.05. The value of the coefficient shall be determined in accordance with 4.5.1 and shall comply with the requirements of Table I (see 6.2).

TABLE I. Grades for noise-reduction coefficient

Grade	NRC
1	0.90 and over
2	0.80 - 0.89
3	0.70 - 0.79
4	0.60 - 0.69
5	0.50 - 0.59
6	0.40 - 0.49
7	**Less than 0.40**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

3.5.2 Sound absorption coefficient. For auditorium treatments, acoustical material specified in the invitation for bids shall be graded in terms of sound-absorbing qualities at 500 Hz. The limiting values of sound-absorption coefficients at 500 Hz for each grade shall be determined in accordance with 4.5.1 and shall comply with the requirements of table II (see 6.2).

TABLE II. Grades for sound-absorption coefficients at 500 Hz (for auditorium treatment)

Grade	Coefficient at 500 Hz
101	0.90 and over
102	0.80 - 0.89
103	0.70 - 0.79
104	0.60 - 0.69
105	0.50 - 0.59
106	0.40 - 0.49
107	Less than 0.40

3.6 Surface-flame spread. When tested in accordance with 4.5.2, acoustical material shall be designated as follows (see 6.2):

3.6.1 Class (a), noncombustible. The flame-spread index of the acoustical material shall not exceed 25 nor shall the material show evidence of continued progressive combustion. Materials with a flame-spread index of 25 or less are considered noncombustible (see 6.2).

3.6.2 Class (b), fire-resistant. The flame-spread index shall not exceed 75 on the exposed surface. Materials with a flame-spread index of 75 or less are considered fire-resistant (see 6.2).

3.6.3 Class (c), slow burning. The flame-spread index shall not exceed 200 on the exposed surface. Material having a flame-spread index of 200 or less is considered slow burning. Materials having a flame-spread index over 200 are considered combustible (see 6.2).

3.7 Dust test. The acoustical material shall be subjected to the dust test specified in 4.5.3. Dust removed from the acoustical material shall not exceed 0.025 gram per square foot. Unless otherwise specified (see 6.2), the dust test shall not be required for civil agency procurement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure **that** supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, examined, and tested to the extent specified, in accordance with the requirements of reference subsidiary specifications and standards. The terms and definitions of MIL-STD-109 shall apply.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one acoustical materials sample when a first article sample is required (see 3.3). This inspection shall include the tests of 4.5. Failure of the first article to pass the tests shall be cause for rejection. The first article may be a standard production item from the contractor's current inventory provided the material meets the requirements of the specification and is representative of the manufacturing technique applicable to the remaining material to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on samples selected in accordance with 4.4 from each lot offered to the Government. The inspection shall include the tests of 4.5.

4.3 Inspection lot. All acoustical material of the same type, density, class, and grade offered for delivery at one time shall be considered a lot for purposes of inspection.

4.4 Sampling for tests.

4.4.1 Sampling for sound absorption test and noise reduction test. Each sample for sound absorption and noise reduction tests shall consist of enough acoustical material, ready for application when mixed with water or binder as applicable to cover an area of approximately 72 square feet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

4.4.2 Sampling for flame-spread test. The sample for flame-spread test shall consist of enough material, ready for application with the possible exception of water only, to cover an area approximately 50 square feet.

4.4.3 Sampling for dust test. The sample for the dust test shall consist of enough material, ready for application with the exception of water, to cover an area of not less than 4 square feet.

4.5 Test methods.

4.5.1 Sound absorption and noise reduction coefficient tests. Sound absorption and noise reduction coefficients of the acoustical sample material shall be determined in accordance with ASTM C423.

4.5.2 Flame-spread index. The flame-spread index shall be determined in accordance with ASTM E84.

4.5.3 Dust test. The dust test sample shall be tested as specified in 4.5.3.2 and 4.5.3.4.

4.5.3.1 Alternate dust test. Upon approval of the contracting officer (see 6.2), an alternate dust test may be substituted for the test described in 4.5.3.2 and 4.5.3.4 provided the requirements of 3.7 have been met.

4.5.3.2 Specimen. The sample of acoustical material in the type and density specified shall be applied to an asbestos-cement sheet conforming to SS-B-755, type F, 3/8-inch thick. Application shall be in strict accordance with the manufacturer's printed instructions, including any prescribed surface preparation and finish treatment, except that not less than 0.5 inch of acoustical material shall be applied. Specimen shall be conditioned at 21° ±1° centigrade (72° ±2° Fahrenheit (F)) and 50 percent relative humidity until the weight of the specimen, weighed at 24-hour intervals, becomes constant to within 0.1 percent.

4.5.3.3 Test apparatus. The test apparatus shall consist of a tight metal duct approximately 17 inches wide by 24 inches high by 7 feet long. A centrifugal blower, driven by a 7.5 horsepower motor, shall be coupled to the duct. One side of the duct shall have an opening large enough to insert a test sample 16 inches wide by 48 inches long, having approximately 4 square feet of surface area exposed to the air flow (see figure 1). A vane anemometer shall be mounted in front of the test panel area for measurement of the air flow velocity. Two dust-collecting filters, having a 95 percent efficiency in filtering particles 5 microns in size and larger, shall be installed upstream of the opening for the test sample (see 6.6). A black dust-collecting

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

cloth shall be placed downstram of the opening for the test sample. The cloth fabric shall be made of 30 denier nylon having a thread count of 102 warp ends per inch and 82.3 filler picks per inch. The cloth shall be dried at 120° ±1°F to a constant weight before each weighing.

4.5.3.4 Procedure. The test for dusting shall be conducted in a reasonably dust-free environment. The test specimen shall be inserted in the test opening in the duct side wall. The complete system shall be purged for 6 hours at a minimum impinging velocity of 800 feet per minute over the entire surface of the test specimen. After the system has been purged for 6 hours, the dust cloth shall be removed and dried at 120°F and weighed until the weight is constant with the nearest 0.001 gram. The dust-collecting cloth shall be reinserted in the test apparatus and the system again purged for 24 hours. The dust-collecting cloth shall then be removed and dried and weighed as before. The net difference in weight between the 6 hour purging and the 24 hour purging shall constitute the dust removed from the acoustical material. The weight of the dust removed shall not exceed the amount specified in 3.7.

4.6 Packaging inspection. An inspection shall be made to determine that the packing and marking comply with the requirements in Section 5 of this specification. Defects shall be scored in accordance with Table III. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping sacks fully prepared for delivery. The lot size shall be the number of shipping sacks in the end item inspection lot. The inspection level shall be S-2 with an Acceptable Quality Level (AQL) of 4.0 defects per hundred units.

TABLE III. Classification of packaging defects

Catagories	Defects
101	Markings omitted, incorrect, or illegible; improper size, location, sequence, or method of application.
102	Any component missing or damaged.
103	Inadequate application of components such as incomplete closure of sacks.
104	Amount per container is more or less than required.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

5.1.1.1 Types I and II. The material shall be furnished in sacks conforming to the level A requirements of UU-S-48 for building plaster as indicated for sack construction number applicable to the weight involved.

5.1.1.2 Type II. The material shall be furnished in 55-gallon tighthead drums conforming to the Uniform Freight Classification or the National Motor Freight Classification.

5.1.2 Level B.

5.1.2.1 Types I and II. The material shall be furnished in sacks conforming to the level B requirements of UU-S-48 for building plaster as indicated for sack construction number applicable to the weight involved.

5.1.2.2 Type III. Same as specified in 5.1.1.2.

5.1.3 Level C.

5.1.3.1 Types I, II, and III. The material shall be packed in a manner which will insure arrival at destination in satisfactory condition condition and be acceptable to the carrier at lowest ratings. Containers and packing shall comply with the Uniform Freight Classification or National Motor Freight Classification.

5.2 Marking. (See 6.2).

5.2.1 Civil agencies. Containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. Containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Acoustical materials are intended for use in the control of reverberation in auditoriums or other rooms. Type I material is normally applied over gypsum and portland cement base coats, and over metal laths and clip-systems. Types II and III materials are intended for spraying beams, roof decks, and columns.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, density, grade, and class of acoustical material required (see 1.2, 3.5 through 3.5.2 and 3.6 through 3.6.3).
(c) Certification of test data if different (see 3.1).
(d) When a first article is required; quantity of material required; and type, density, grade, and class required (see 3.3).
(e) If dust test is required by civil agencies (see 3.7).
(f) When alternate dust test is acceptable (see 4.5.3.1).
(g) Selection of applicable level of packing required (see 5.1).
(h) Marking required (see 5.2).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulation (ASPR). The first article may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one quantity of each classification as specified. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examination, tests, and approval of the first article.

6.4 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the ASPR, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.5 Application of materials. To insure proper and uniform thickness, it is recommended that acoustical materials be applied between spot ground or proper size, spaced at intervals of about 8 feet, or by other means recommended by the manufacturer to give equivalent results. After application, the acoustical material should be finished to a true surface with a cork float, steel trowel, roller, or other means.

6.6 Paintability. When acoustical material covered by this specification is to be painted, a nonbridging paint should be used to prevent damage to the sound-absorption property.

6.7 Filters. The Farr HP-2A filter, used as part of the test apparatus, is manufactured by the Farr Company, P.O. Box 90187, Airport Station, Los Angeles, CA 90009 and is suitable for filtering particles as small as 5 microns.

6.8 Classification changes. Changes in classification of the acoustical materials between this version of the specification and the previous edition are as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

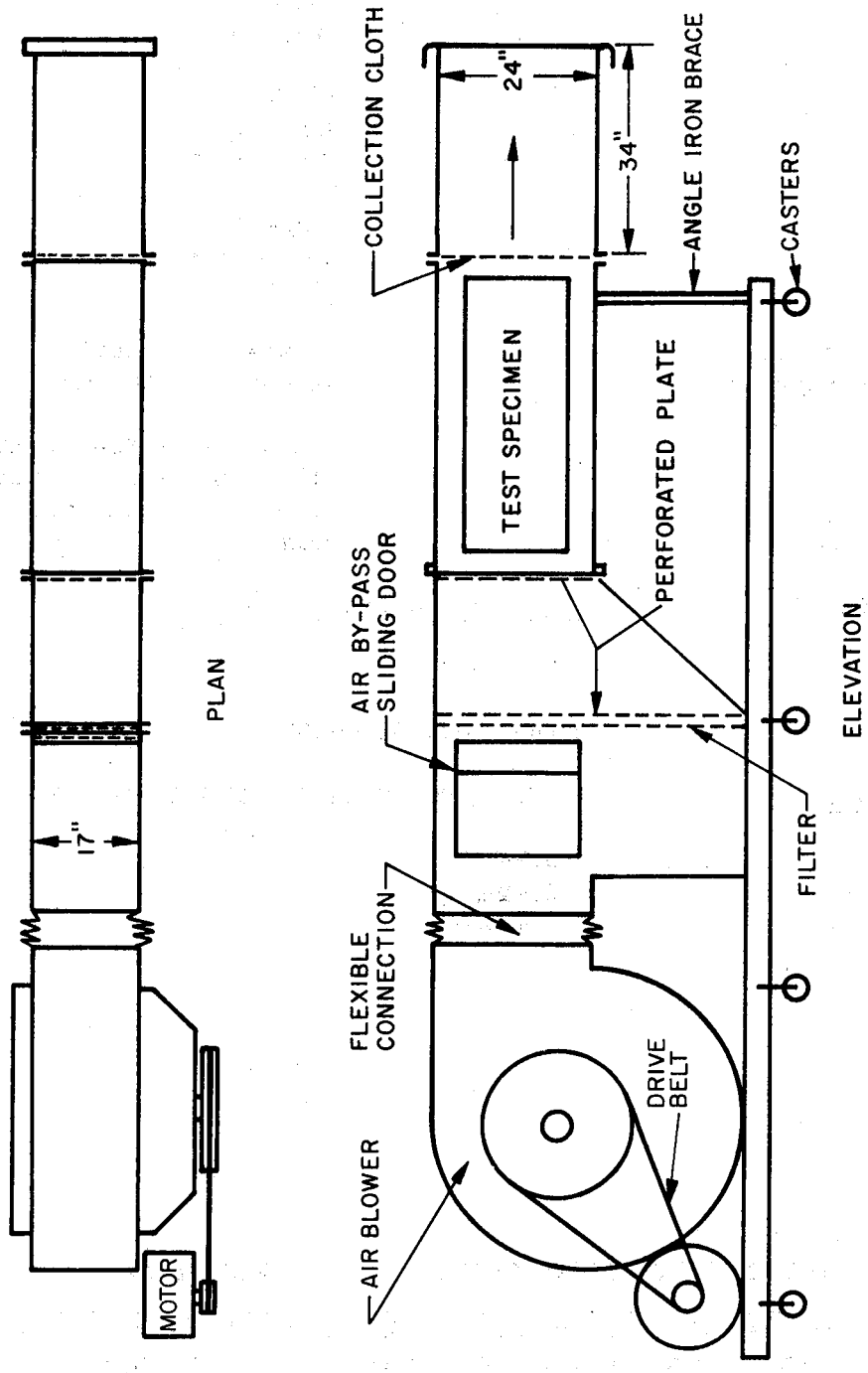


FIGURE I. TYPICAL APPARATUS FOR DUST TEST

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.

Old Designation	New Designation
Type I	Type I
Density A	Density A
Density B	Density B
Type II	Type II
Density C	Density C
Density D	Density D
None	Type III
None	Density E
None	Density F
General description	Grades 1 through 7
General description	Grades 101 through 107
Class 25	Class (a) - Noncombustible
Class 75	Class (b) - Fire-resistant
Class 200	Class (c) - Slow burning
Class over 200	None

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 84

Review activities:

Army - CE
Air Force - 84

User activities:

Army - ME

Preparing activity:

Navy - YD

Project No. 5610-0077

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
VA-DMS
COM-NBS

OOrders for this publication are to be placed with General Services Administration, acting as agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. **Price _ cents each.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:12Z
Check the source to verify that this is the current version before use.